```
while True:
    ret, frame = video_capture.read()
    rgb_frame = frame[:, :, ::-1]

    # Detect Faces
    face_locations = face_recognition.face_locations(rgb_frame) face_encodings =
face_recognition.face_encodings(rgb_frame, face_locations) # Match Faces
    for (top, right, bottom, left), face_encoding in zip(face_locations, face_encodings): matches =
face_recognition.compare_faces(known_face_encodings, face_encoding) name = "Unknown"

        # Use the first match
        if True in matches:
            match_index = matches.index(True)
            name = known_face_names[match_index]

        # Display Label
        cv2.rectangle(frame, (left, top), (right, bottom), (0, 255, 0), 2) cv2.putText(frame, name, (left,
top - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.9, (0, 255, 0), 2) # Show Frame
    cv2.imshow('Video', frame)

    # Exit on 'q'
    if cv2.waitKey(1) & 0xFF == ord('q'): break

# Release Resources
video_capture.release()
cv2.destroyAllWindows()
```

Project 3: Sentiment Analysis for Customer Feedback

Objective:

To analyze customer reviews and feedback to identify sentiments as positive, neutral, or negative. The project aims to empower businesses to understand customer opinions at scale and prioritize improvements effectively.

Detailed Plan:

1. Data Collection:
    - Gather customer reviews from platforms such as Amazon, Twitter, or company websites.
    - Use web scraping tools like BeautifulSoup or Scrapy to extract review data if APIs are unavailable.